

① $X_1, \dots, X_N$

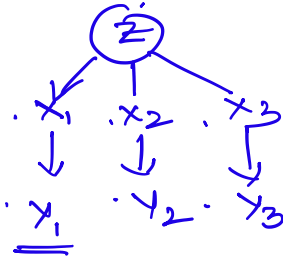
$$Q \subseteq \{X_1, \dots, X_N\}$$

$$E \subseteq \dots$$

$$H \subseteq \dots$$

$$P(Q | e_1 \dots e_k) = \frac{\sum_{h \in H} P(Q, e_1 \dots e_k, h_1 \dots h_L)}{\sum_{h \in H} P(Q, e_1 \dots e_k, h_1 \dots h_L)}$$

enumeration.



$$P(X_1, X_2, X_3, Y_1, Y_2, Y_3 | Z)$$

$$= P(Y_1 | X_1) P(X_1 | Z) P(Y_2 | X_2) P(X_2 | Z) P(Y_3 | X_3) P(X_3 | Z) P(Z)$$

$$V, E = \{Z, X_1, X_2\}$$

$$P(X_3 | Y_1, Y_2, Y_3) \quad P(X_3 = x_3 | Y_1 = y_1, Y_2 = y_2, Y_3 = y_3)$$

01 X_1, X_2, Z

$$P(Y_1 = y_1 | X_1) \quad P(Y_2 = y_2 | X_2) \quad P(Y_3 = y_3 | X_3)$$

$$P(y_1 | x_1) \quad P(y_2 | x_2) \quad P(y_3 | x_3)$$

$$P(y_1 | \neg x_1) \quad P(y_2 | \neg x_2) \quad P(y_3 | \neg x_3)$$

Join on X_1

$$P(X_1 | Z) \quad P(Y_i = y_i | X_1)$$

$$\text{Eliminate } X_1 \quad f_{X_1}(y_1 | Z)$$

Z	X ₁
z	x ₁
z	¬x ₁
¬z	x ₁
¬z	¬x ₁

X ₁	Y ₁
x ₁	y ₁
¬x ₁	y ₁

$$\frac{x_1 y_1 z}{x_1 y_1 \neg z} \quad \frac{\neg x_1 y_1 z}{\neg x_1 y_1 \neg z}$$

$$y_1 z \quad y_1 \neg z$$

Join on X_2

Eliminate on X_2

$$f_{X_2}(y_2 | Z)$$

X ₂	Y ₂
x ₂	y ₂
x ₂	¬y ₂
¬x ₂	y ₂
¬x ₂	¬y ₂

Y ₂	Z
y ₂	z
¬y ₂	¬z

$$z \quad \neg z$$

Join on X_3 $f(X_3, Z, y_3)$

Join on Z .

$$P(Z) f_{x_1}(y_1|Z) f_{x_2}(y_2|Z) f(x_3, Z, y_3)$$

$y_1 \ y_2 \ y_3 \ x_3 \ Z$ elimination Z (2)

$$y_1 \ y_2 \ y_3 \ x_3 \ Z \checkmark$$

$$\neg x_3 \ Z \times$$

$$x_3 \neg Z \checkmark$$

"

$$\neg x_3 \neg Z \times$$

$$y_1 \ y_2 \ y_3 \ x_3$$

$$y_1 \ y_2 \ y_3 \ x_3$$

"

$$\neg x_3$$

$$P(x_3, y_1, y_2, y_3)$$

$$Z, x_1, x_2 \ P(x_3 | y_1, y_2, y_3)$$

Solve Z

$$P(Z) P(x_1|Z) P(x_2|Z) P(x_3|Z)$$

$$\textcircled{Z} \quad \textcircled{x_1 \ Z} \quad \textcircled{x_2 \ Z} \quad \textcircled{x_3 \ Z}$$

$$\neg Z$$

$$x_1 \neg Z$$

$$\neg x_1 \ Z$$

$$\neg x_1 \neg Z$$

$$\neg x_2 \ Z$$

$$x_1 \quad x_2 \quad x_3 \quad Z$$

$$x_1 \quad x_2 \quad x_3 \quad Z$$

$$x_1 \quad \neg x_2 \quad x_3 \quad Z$$

$$x_1 \quad x_2 \quad \neg x_3 \quad Z$$

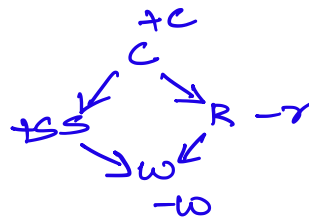
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Eliminate on Z $f_Z(x_1 x_2 x_3)$

$$x_1 \ x_2 \ x_3 \quad \underline{\underline{8}}$$

likelihood weighting:

$$P(C|\underline{+S})$$



$$w_1 = P(+S|C) \leftarrow \frac{+C + S - R - W}{P(C) P(R|C) P(W|S, R)}$$

$$w_2 = P(+S|-C) = -C + S + R + W$$

$$\prod_{i=1}^K P(e_i | \text{Parents}(e_i))$$

likelihood weight
($x_1 \dots x_N$)

$$P(C|\underline{+S})$$

learning: